MINH KHUE LE

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EDUCATION

Hanoi University of Science and Technology Hanoi, Vietnam

B.Eng. in Control Engineering and Automation, GPA 3.42/4.0 *May 2023*

- Participated in undergraduate research under the guidance of Assoc. Prof. Dao Quy Thinh, resulting in an undergraduate thesis on autonomous mobile robot control and local path planning algorithms.
- Thesis title: *Development of a Controller for Autonomous Mobile Robots and Local Path Planning Algorithm*

RESEARCH EXPERIENCE

Welfare and Service Robotics Laboratory Hanoi, Vietnam

Hanoi University of Science and Technology

Advisor: Assoc. Prof. Dao Quy Thinh

My research focuses on control systems and navigation algorithms for autonomous mobile robots (AMR), with an emphasis on local path planning in dynamic environments.

- Developed control algorithms for autonomous mobile robots, ensuring stable motion and trajectory tracking
- Designed and implemented local path planning algorithms for real-time navigation and obstacle avoidance
- Integrated control and planning modules in [ROS / simulation / real robot platform]
- Evaluated system performance in [simulation/real-world experiments], improving navigation efficiency and robustness

Computer-Integrated Manufacturing Laboratory Hanoi, Vietnam

Hanoi University of Science and Technology

Advisor: Assoc. Prof. Nguyen Ngoc Kien

Participated in the research and development of an automated bolt–nut assembly machine for industrial assembly lines.

- Developed an automated feeding and sorting mechanism for bolts, nuts, spring washers, and flat washers
- Designed a passive mechanical guiding mechanism for automatic assembly without the use of motors
- Achieves a tightening accuracy within ±1 thread turn without using sensors
- Contributed to system integration combining mechanical design and control logic for reliable operation

Intelligent and Optimal Systems Laboratory Hanoi, Vietnam

Hanoi University of Science and Technology

Advisor: Assoc. Prof. Nguyen Van Tinh

My research focuses on robotic automation systems for smart kitchen applications, including task execution, system integration, and control of electromechanical components.

- Designed and implemented task planning strategies for sequential food preparation operations
- Integrated hardware and software modules (actuators, sensors, controllers) into an autonomous kitchen system
- Implemented system control and coordination using C# and web-based interfaces.
- Evaluated system performance through real-world experiments, improving reliability and operational efficiency

RESEARCH INTEREST

- Control and Navigation for Autonomous Mobile Robots
- Multi-Robot Coordination and Task Allocation
- AI-based Perception and Decision-making for Robotics
- Learning-based Control and Motion Planning

AWARDS AND HONORS

- **Viettel Excellence Scholarship (2025)** — Fully funded graduate study in Robotics
- First Prize, **Canon ChieTech Technology Intelligence Competition (2021)**
- Second Prize, **Panasonic IoT Innovation Idea Contest (2018)**
- Top 20 Finalist, **SIMATIC IoT 2050 Competition**

INDUSTRY EXPERIENCE

Viettel Post Corporation (Viettel Group) May 2023 – Present

R&D Engineer, Robotics Department *Hanoi, Vietnam*

- Developed traffic coordination and task allocation algorithms for AGV systems, along with communication protocols for multi-robot coordination
- Designed scalable fleet management architecture supporting up to 200 robots
- Designed and implemented control and coordination software for AMR picking systems and autonomous forklifts
- Integrated heterogeneous mobile robots from external vendors (e.g., iRayple-Dahua) into unified control systems
- Led end-to-end development of logistics robot projects (towing, picking, and CTU systems), covering system architecture, mechanical and electrical design, firmware, and control software
- Developed autonomous delivery robot solutions for last-mile logistics applications
- Implemented autonomous container transportation systems at cross-border logistics hubs (Lang Son – Guangxi, China; Lao Cai – Hekou, China)
- Authored patented and proprietary technologies in AMR systems, including:
    - Deep learning-based vehicle detection and localization
    - Autonomous charging systems for towing AMRs
    - Natural environment-based localization methods
    - Fleet control and coordination algorithms
    - Software copyright: robot fleet management and coordination systems (granted)
- Selected for the **Viettel Excellence Scholarship (2025)**, awarded to top engineering talents

PAC Automation JSC August 2022 – April 2023

Lead R&D Engineer *Hanoi, Vietnam*

- Modified and integrated control systems for automated kitchen equipment (robots, mixers, conveyors)
- Developed control software and customized imported automation systems
- Collaborated with cross-functional teams to deploy integrated solutions

PUBLICATIONS

(Accepted)

To Minh, H.*, Ngo Thi Thuy, D., Pham Tien, D., Phung Van, H., Le Minh, K., Nguyen Kim, T., Giang Quoc, H., Ha Thanh, S., Nguyen Van, T. Graduated Dwell-Time Hysteresis for Mode-Switching Stability in Multi-Modal Mobile Robots. *Robotics and Mechatronics: Proceedings of ISRM 2026*.

(In Preparation)

Stability-Aware Hybrid A* Path Planning for Multi-Modal Four-Wheel

Steering Robots in Confined Dead-End Environments

PATENTS AND INTELLECTUAL PROPERTY

Software Copyrights (Granted):

- Robot Picking Management, Coordination, and Monitoring Software

Patents and Applications (Filed, expected Dec 2027):

- Method for vehicle detection and localization using deep learning
- System and method for automatic charging of towing AMRs
- Localization method for towing AMRs based on natural environment features
- Control and coordination solution for towing AMR systems

Industrial Design (Filed):

- Industrial design of towing Autonomous Mobile Robot (AMR)

Software Copyrights (Filed):

- AMR Fleet Management, Monitoring, and Coordination Software

Patents and Works (In Preparation):

- Industrial mechanical design for hybrid indoor–outdoor delivery AGV systems
- Method for dynamic obstacle detection and trajectory prediction using camera–LiDAR fusion
- Autonomous navigation system for delivery AMRs with real-time environment understanding
- Adaptive driving mode switching method for AMRs under varying conditions
- Fleet coordination methods for autonomous delivery robots

Industrial Design (In Preparation):

- Industrial design of outdoor autonomous delivery AMR

Software Copyrights (In Preparation):

- Autonomous Delivery Vehicle Management and Coordination Software

SKILLS

Languages: Vietnamese (native), English (working proficiency)
Programming & Tools: Python, C#, MATLAB, ROS

- Multi-robot coordination, control algorithms, simulation

Software & Systems: Java, basic web development (HTML, CSS, JavaScript)
Engineering Software: SolidWorks, AutoCAD
Automation & Control: PLC (Siemens, Mitsubishi, Omron, Delta, Weitek), HMI

- Industrial automation systems, system integration

REFERENCES

Assoc. Prof. Dao Quy Thinh
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Hanoi University of Science and Technology
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Assoc. Prof. Nguyen Ngoc Kien
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